

On the **compatibility** of
dependent type theory
with Brouwer's **Intuitionistic** principles



Maria Emilia Maietti

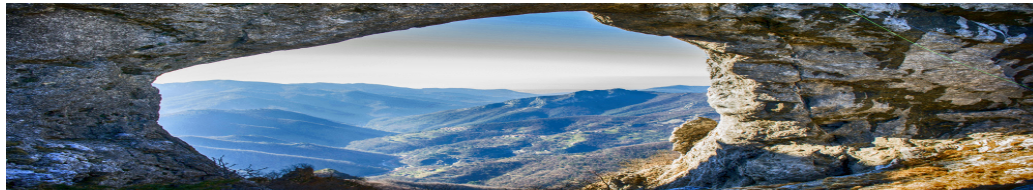
University of Padova

Types and Topology, **Birmingham, 17-18/912/25**

A workshop in honor of Martin Escardò 's 60th birthday

Abstract

- Bishop's **Compatibility** issues about **constructive maths**
- **Compatibility**/Incompatibility with Brouwer's **Intuitionistic** maths
of **type theoretic** foundations for **constructive maths**
- **Open problem** about **Brouwer's intuitionistic** principles

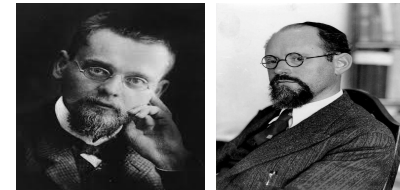


Brouwer's intuitionistic Mathematics

intuitionistic mathematics by Brouwer



is Incompatible with classical mathematics



because of Brouwer's continuity principles

implying that

all functions between real numbers must be continuous!

Brouwer's intuitionistic Mathematics



Brouwer's intuitionistic continuity principles:

Bar Induction (BI) = spatiality of Baire locale

($\Rightarrow$ we can reason inductively in the Baire space)

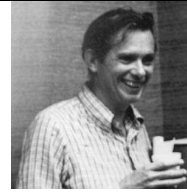
Local Continuity Principle (LCP) = continuity of all functions from Baire space to Nat

where

Brouwer's choice sequences = functional relations

Constructive Mathematics between intuitionistic and classical one

according to Errett Bishop



author of "Foundations of Constructive Analysis" 1967

constructive maths must be compatible

both with classical and intuitionistic mathematics

constructive maths

classical

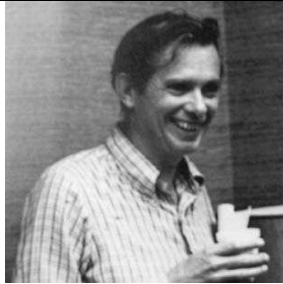


intuitionistic



Constructive Mathematics by Bishop

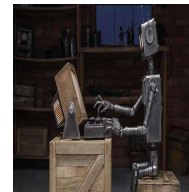
Bishop



showed in "Foundations of constructive analysis"
that a large portion of functional analysis
can be reproduced **constructively**

with **constructive proofs**
having a **computational contents**
so that the **existence** of an **object**
can be **computed** by a **machine**

and **axiom of choice** is valid

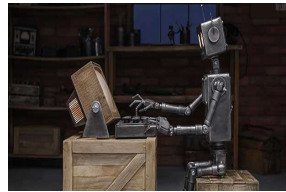


Essence of **constructive** mathematics

constructive mathematics

=

IMPLICIT **computational** mathematics



made **EXPLICIT**

in **computable** models

validating **Church thesis CT**

+

Axiom of choice AC

for **program**-extraction from **proofs**

at least on **Natural Numbers**

A principle of recursive mathematics: Church Thesis



$$\begin{aligned} (\text{CT}) \quad & \forall x \in \text{Nat} \exists! y \in \text{Nat} \ R(x, y) \\ & \exists e \in \text{Nat} \quad (\forall x \in \text{Nat} \exists y \in \text{Nat} \ T(e, x, y) \ \& \ R(x, U(y))) \end{aligned}$$

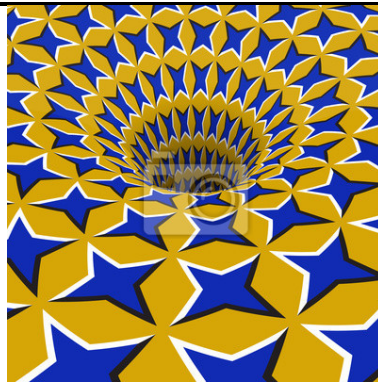
functional relations are all **computable**

valid in recursive constructive maths



of Russian constructivism initiated by Markov

Number-theoretic Axiom of choice



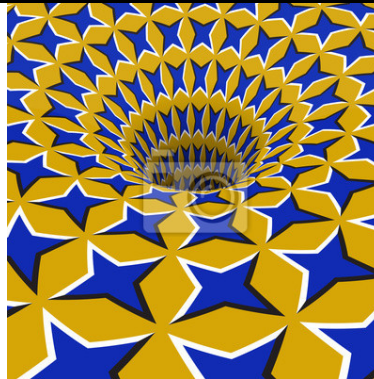
$(AC_{Nat, Nat})$

$\forall x \in Nat \exists y \in Nat R(x, y) \longrightarrow$

$\exists f \in Nat \rightarrow Nat \forall x \in Nat R(x, f(x))$

from any **total relation** we can extract a function

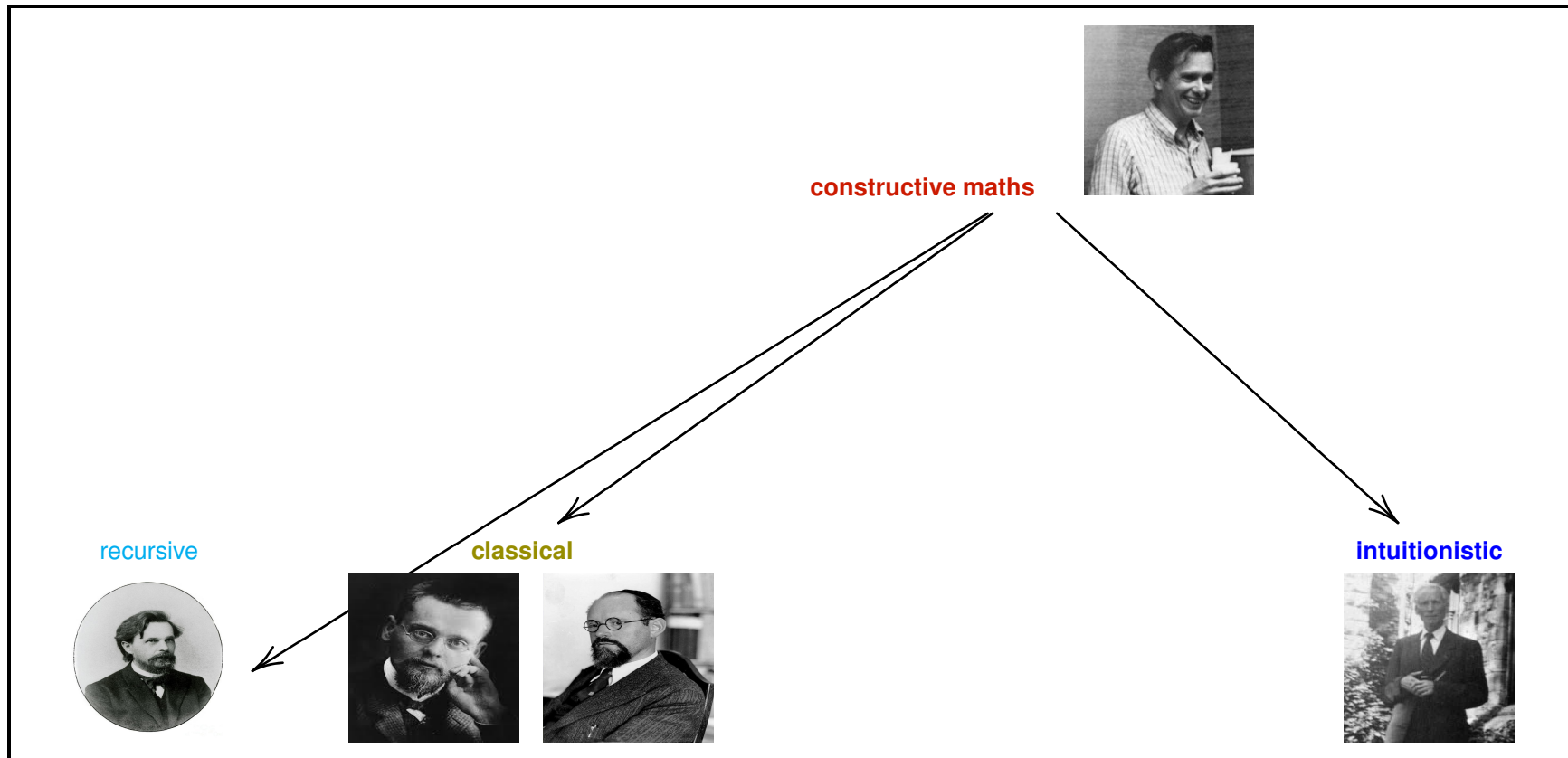
Axiom of choice



$$(\textcolor{violet}{AC}) \quad \forall x \in A \quad \exists y \in B \textcolor{blue}{R}(x, y) \longrightarrow \exists \textcolor{red}{f} \in A \rightarrow \textcolor{red}{B} \quad \forall x \in A \quad \textcolor{blue}{R}(x, \textcolor{red}{f}(x))$$

from any **total relation** we can extract a function

Refinement: **Constructive** Maths among recursive, intuitionistic and classical ones



Brouwer's intuitionism incompatible with Russian constructivism



Brouwer's continuity principles + Church thesis $\vdash \perp$

($\Rightarrow$ incompatible with constructive recursive maths
of Russian constructivism including CT)

i.e. with because

$HA^\omega + \text{Fan theorem} + \text{CT}$ is inconsistent

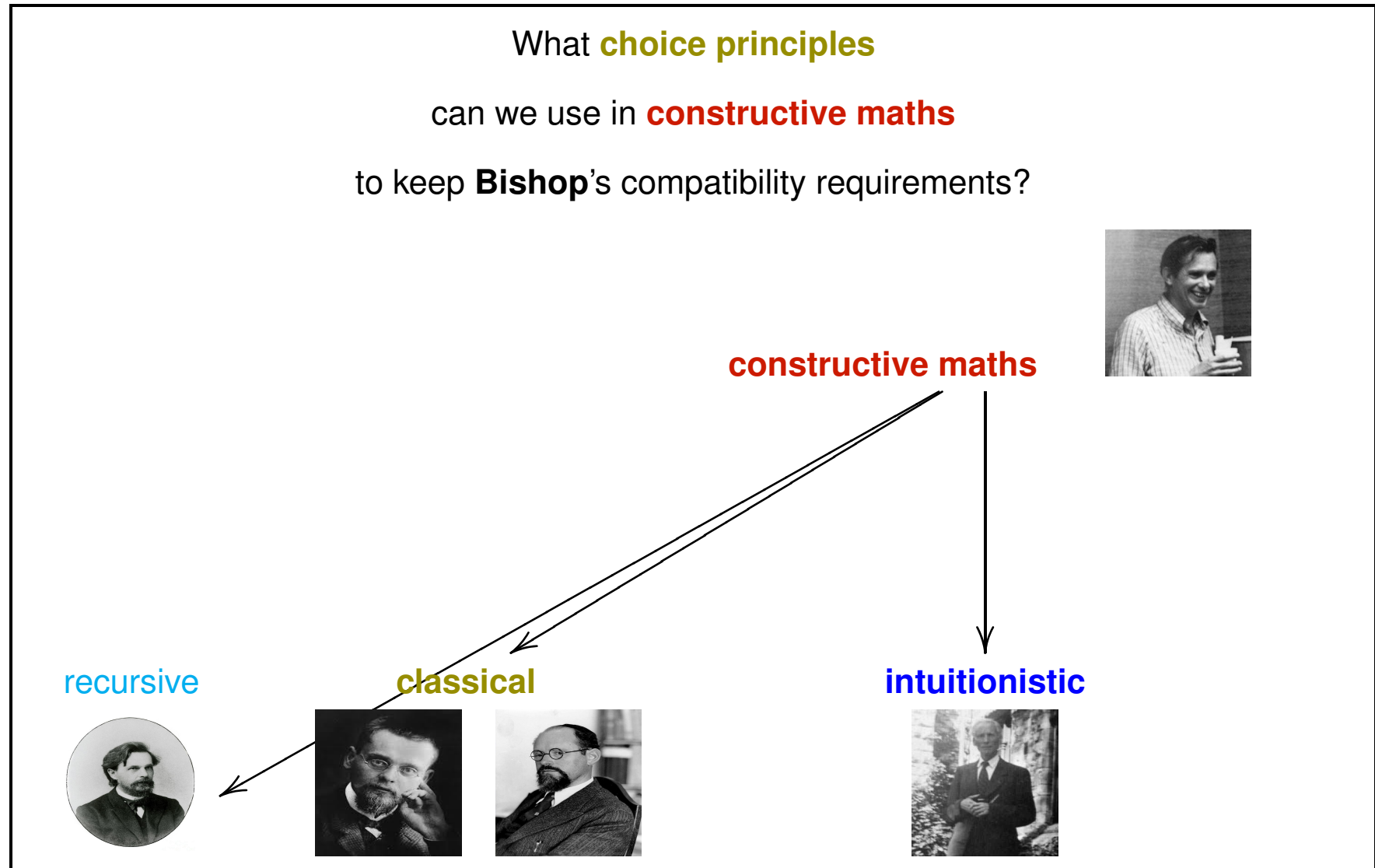
where

Fan theorem = *spatiality* of Cantor space

consequence of BI

choice sequences = functional relations

What choice principles in Constructive Maths ?



Brouwer's LCP contradicts Axiom of Choice (AC) and extensionality



from Troelstra 1977

Heyting arithmetics with finite types + LCP + AC+ extfun $\vdash \perp$

where

$$\text{extfun} \frac{f(x) =_B g(x) \text{ true } [x \in A]}{\lambda x. f(x) =_{A \rightarrow B} \lambda x. g(x) \text{ true}} \quad \begin{array}{l} \text{extensionality} \\ \text{of functions} \end{array}$$

Brouwer's LCP contradicts Axiom of Choice (AC) contradicts the ξ -rule



from Escardò-Xu 2015

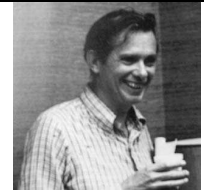
Martin-Löf's type theory + **LCP** + **AC** + **ξ -rule** $\vdash \perp$

where

$$\xi\text{-rule} \frac{f(x) = g(x) \in B(x) \quad [x \in A]}{\lambda x. f(x) = \lambda x. g(x) \in \Pi_{x \in A} B(x)}$$

What foundation for constructive mathematics ??

Since the 80s various foundations for Bishop's constructive mathematics appeared including



Martin-Löf's type theory - incompatible with **Brouwer's intuitionistic maths**

Aczel set theory - incompatible with **CT** + **axiom of choice**

Homotopy type theory - incompatible with **CT** + **axiom of choice**

...



Church Thesis CT + Axiom of Choice AC contradicts extensionality



Heyting arithmetics with finite types + CT + AC + extfun $\vdash \perp$

Why a **minimalist foundation** for **constructive mathematics** ??

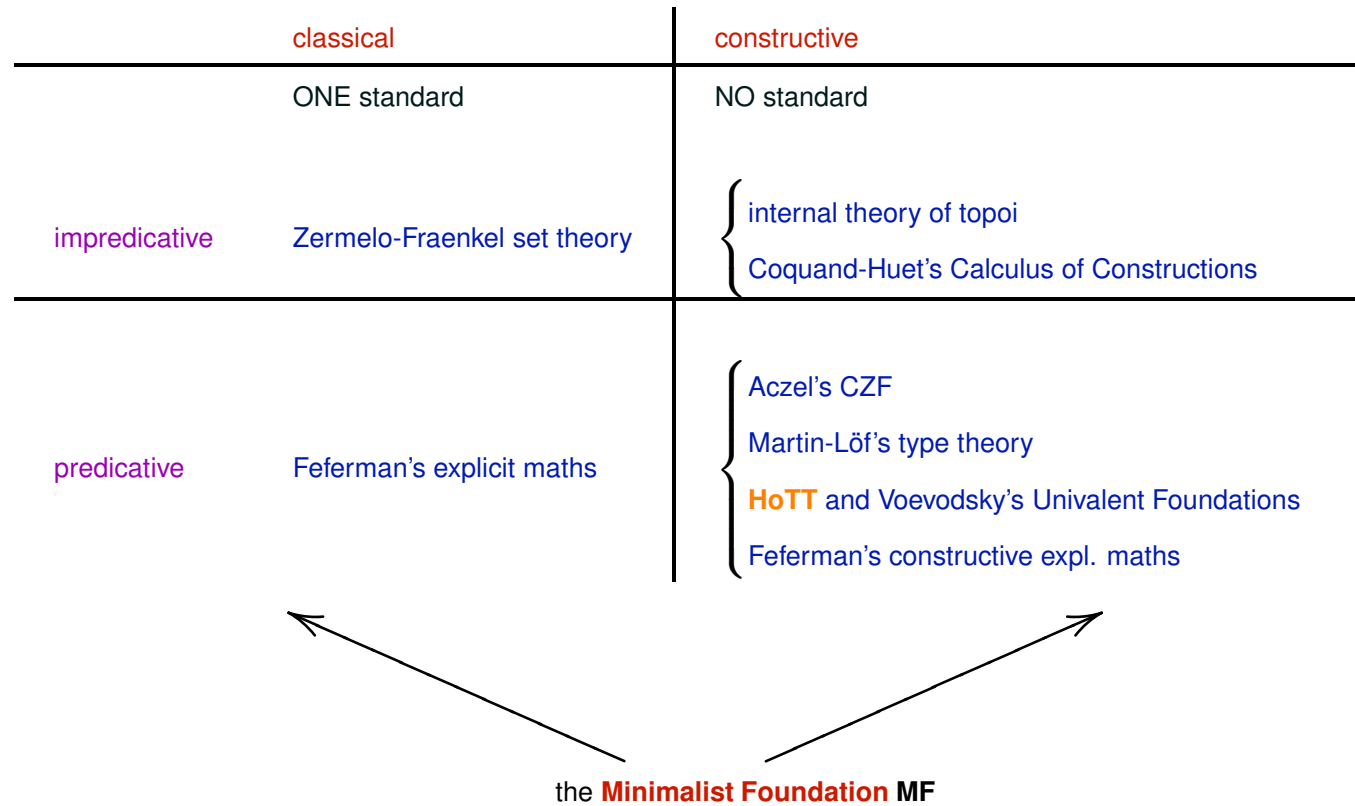
j.w.w. Giovanni Sambin

We wanted to take advantage of the **plurality**

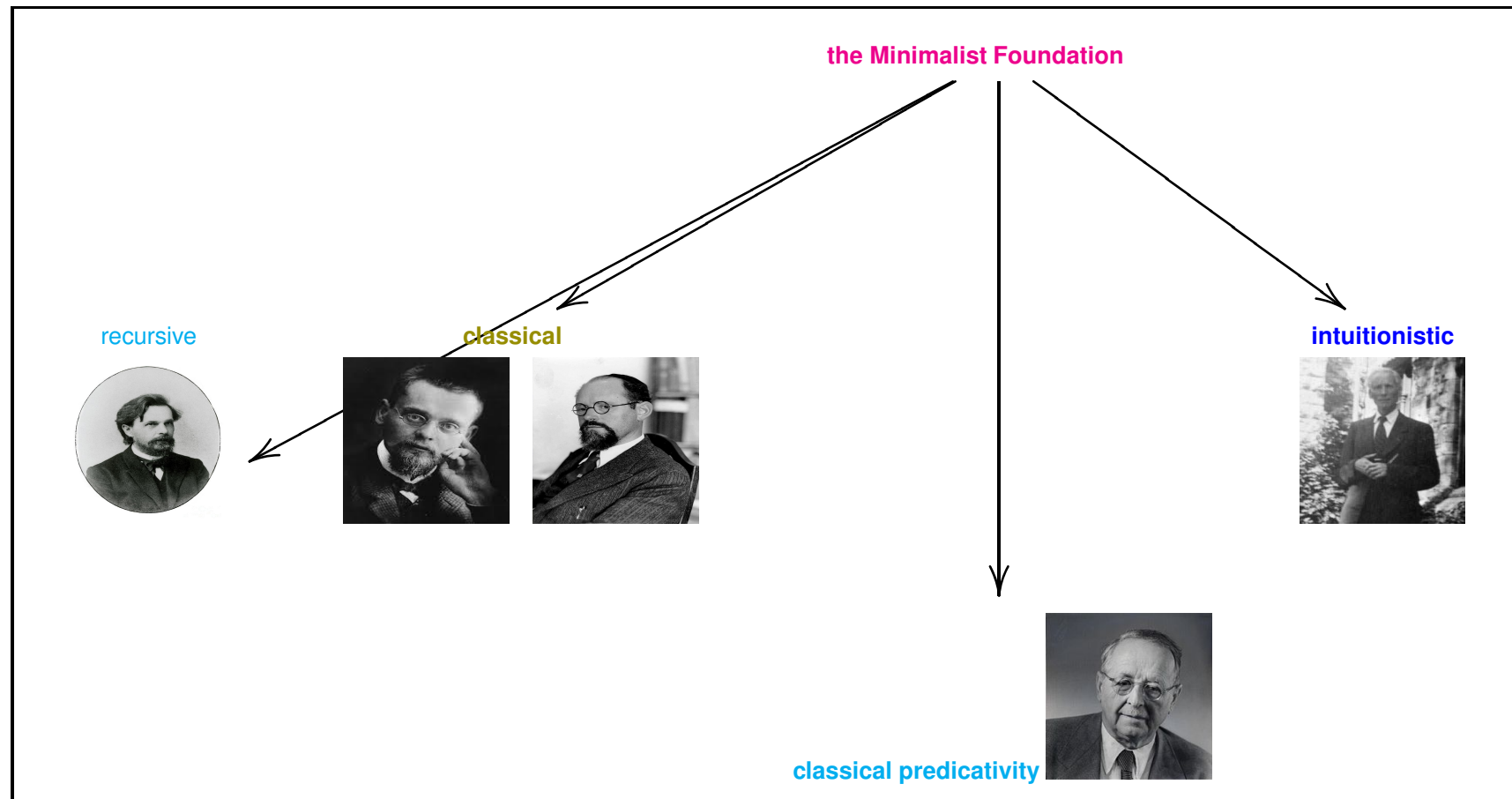
of **foundations** for **Bishop's constructive mathematics**



*Plurality of foundations $\Rightarrow$ need of a **minimalist foundation***



Peculiarity of the Minimalist Foundation for constructive maths



Our TWO-LEVEL Minimalist Foundation

from [Maietti'09] in agreement with [M. Sambin2005]

its **intensional** level **mTT**



(**Minimalist Type Theory**)

= a **PREDICATIVE VERSION** of **Coquand-Huet's Calculus of Constructions CC**

(fragment of **Rocq**)

(consistent with **CT** + **AC**)

its **extensional** level **emTT**

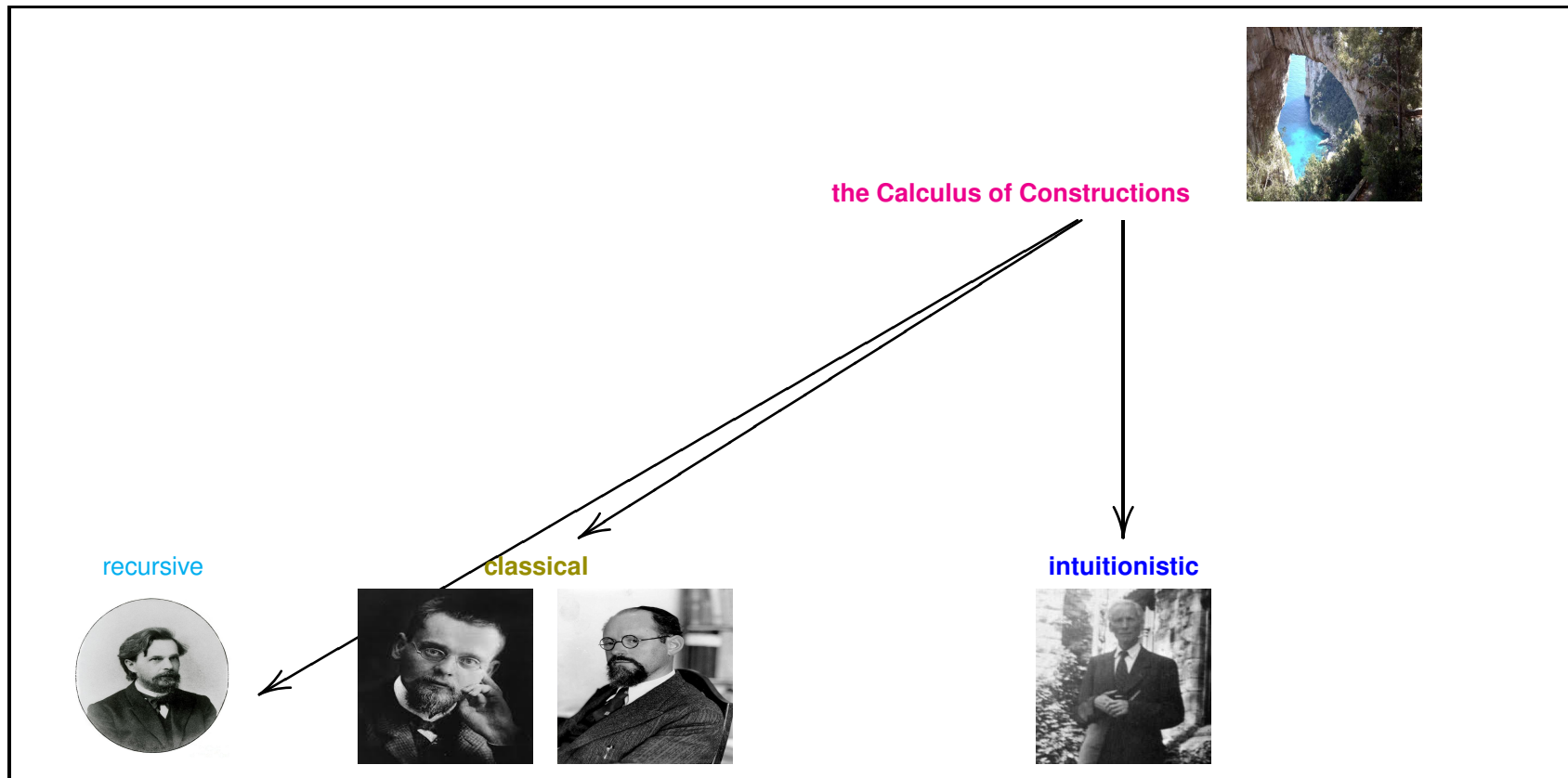


(**extensional Minimalist Type Theory**)

is a **PREDICATIVE LOCAL** set theory

(**NO** choice principles)

Peculiarity of the Calculus of Constructions for Rocq



two notions of function in the Minimalist Foundation



a *primitive notion* of **type-theoretic function**

$$f(x) \in B \ [x \in A]$$

(**closed under "exponentiation"**)

$\neq$ (syntactically)

notion of **functional relation**

$$\forall x \in A \ \exists! y \in B \ R(x, y)$$

(**NOT closed under "exponentiation"**)

Axiom of unique choice



$$\forall x \in A \exists! y \in B R(x, y) \longrightarrow \exists f \in A \rightarrow B \forall x \in A R(x, f(x))$$

turns a **functional relation** into a **type-theoretic function**.

$\Rightarrow$ **identifies** the **two** distinct notions...

Type-theoretic Church thesis



(TCT)

$$\forall f \in \text{Nat} \rightarrow \text{Nat} \quad \exists e \in \text{Nat}$$

$$(\forall x \in \text{Nat} \quad \exists y \in \text{Nat} \quad T(e, x, y) \ \& \ U(y) =_{\text{Nat}} f(x))$$

type-theoretic functions ($\subsetneq$ functional relations)

are all **computable**

Peculiarity of MF: reconciling Russian constructivism with Brouwer intuitionism

Theorem:

Both levels of MF are consistent with + Theoretical Church Thesis (TCT)

+ Brouwer's continuity principles

+ Zermelo's AC_{Baire}^{Nat} (extracting a functional relation)

Bar Induction (BI) = spatiality of Baire locale

Local Continuity Principle (LCP) = continuity of functions from Baire space to Nat

where

Brouwer's choice sequences = functional relations



Proof: a model made of Hyland's assemblies formalized in

Aczel's CZF+REA + BI +LCP

(= a predicative and constructive quasi-topos of assemblies!!!)

Decomposition of Church Thesis



CT
$\Updownarrow$
$\text{TCT} + \text{AC!}_{Nat, Nat}$

Axiom of unique choice



$$\forall x \in A \exists! y \in B R(x, y) \longrightarrow \exists f \in A \rightarrow B \forall x \in A R(x, f(x))$$

turns a **functional relation** into a **type-theoretic function**.

$\Rightarrow$ **identifies** the **two** distinct notions...

valid in Homotopy Type Theory, **Martin-Löf's type theory** and **Aczel's CZF**

but NOT in **the Minimalist Foundation**

Peculiarity of CC: reconciling Russian constructivism with Brouwer intuitionism

Theorem:

CC is consistent with + Theoretical Church Thesis (TCT)

+ Brouwer's continuity principles BI + LCP

+ Zermelo's AC_{Baire}^{Nat} (extracting a functional relation)

where

Brouwer's choice sequences = functional relations

Proof. a model is given by Hyland's assemblies formalized in

IZF + BI + LCP

(= an intuitionistic quasi-topos of assemblies!!!)

thanks to

the equiconsistency of CC with its extensional version in [M. & Sabelli 2024]

which is the internal language of arithmetic solid quasi-toposes



Open issues



from Troelstra 1977 we know that

$HA^\omega + AC + BI + LCP$ is consistent

+

Escardò-Xu's incompatibility of the ξ -rule with intuitionistic maths



- Is the intensional level of the Minimalist Foundation (which does not have the ξ -rule) consistent with $AC + BI + LCP$??
- Is Coquand-Huet's Calculus of Constructions with no ξ -rule consistent with $AC + BI + LCP$??